

UNIT 3: Weather Chart Analysis

c. Mesoscale Analysis

Introduction to Mesoscale Analysis

Mesoscale features are small, localized weather patterns typically spanning less than 650 miles and often centered around individual storms. This objective is designed to help you develop the skills necessary to perform a basic mesoscale analysis. Through lectures, guided tasks, and targeted questions, you will learn to identify and label various weather features using foundational analysis techniques. One essential tool in this process is station plot data, which you encountered earlier in the unit. These plots use specific symbols to represent weather conditions reported by stations, and certain symbols are especially useful for identifying mesoscale systems and the areas they impact. Throughout this objective, you'll explore different mesoscale weather systems and learn how to interpret the symbols that represent them. A Progress Check (PC) at the end of the objective will assess your competency and passing it is required to advance to the next stage. Be sure to follow your instructor's guidance closely to meet all requirements and successfully complete this objective.

Thunderstorm Atmospheric Ingredients

Many of the most severe weather features will originate from strong thunderstorms. For a thunderstorm to develop, there are certain atmospheric dynamics that are required to produce and maintain a thunderstorm. The main "ingredients" that are needed are (MILE):

(M)Moisture

All thunderstorms require moisture. Moisture sources include large bodies of water such as the oceans or large lakes. You learned previously how to analyze moisture and moist air advection. You will need to analyze the synoptic level flow influencing the area. If the overall flow upstream of your location is coming from a moisture source, then you have the first ingredient needed for thunderstorm development.

(I)Instability

The next ingredient needed is an unstable atmosphere. For thunderstorms to develop, you need a favorable environment that enhances upward vertical motion. The more unstable the atmosphere, the easier air parcels rise. Inversely, the more stable the atmosphere, the more upward vertical motion is restricted, and the air parcels will not be able to rise to the height required for thunderstorms to develop.

Cold & Warm Air Advection: One of the causes of an unstable atmosphere is cold air over warm air. If you have an increase in cold air advection aloft, or warm air advection at the surface, you would be increasing the temperature contrast between the upper and lower atmosphere, which would cause the atmosphere to become more unstable.

Daytime Heating & Nighttime Cooling: Daytime heating can also cause more instability due to the surface temperatures rising which would increase the temperature contrast between the surface air and the now relatively colder air above it. The opposite would also be true, that nighttime cooling would increase the cold air near the surface and would cause stability to increase.

(L)Lift

If air parcels are rising, then a lifting mechanism is present. Assuming you have sufficient moisture and an unstable atmosphere, there must be something to lift the parcels off the surface and will be enhanced by an unstable environment. In weather, there are two categories of lifting mechanism that affect rising air parcels:

Free Convection: Free convection describes air parcels being lifted from the surface due to daytime heating. As the surface temperature increases, the air parcel is lifted from the surface and begin cooling adiabatically. It is called free convection because there are no weather features affecting the upward vertical motion of the air parcel.

Forced Convection: Forced convection describes air parcels that are lifted into the atmosphere due to the influence of surrounding weather features. Forced convection occurs when something forces convergence at the surface like a frontal boundary, or something forcing divergence aloft like a Major Short-Wave Trough.

The strength of the lifting mechanism and the amount of instability are closely related, and one can impact the other. The more unstable the environment, the less lift you need. Alternately, the more stable your environment, the stronger your lift needs to be to develop thunderstorms.

(E)Exhaust

The last ingredient needed to maintain and strengthen a thunderstorm is exhaust. With instability and lift, all that energy and upward vertical motion must go somewhere. If there is nothing providing exhaust within a storm, a thunderstorm may develop, but it will eventually collapse in on itself.

If the thunderstorm is to continue increasing in size and intensity, there must be a form of exhaust in the upper levels to allow the excess rising energy somewhere to go to maintain the constant flow of upward vertical motion. We can analyze exhaust by looking for areas of divergence or strong wind bands of 40KT or more at 500mb near the developing thunderstorm.

Exercise / Practice: M.I.L.E

Thunderstorm Types and Intensities

Thunderstorm Types

Single Cell: Single cell thunderstorms, also called air mass thunderstorms, are formed due to free convection. They are short-lived storms usually lasting 30 to 60 minutes and potential severe conditions include high wind speeds and small to medium-sized hail.

Multicellular

There are two types of multicellular thunderstorms: Mesoscale Convective Complexes (MCC) and Mesoscale Convective Systems (MCS). The average duration for both types is 6 to 12 hours, and they form in regions of strong directional shear in the lower levels with strong speed shear aloft. These types of thunderstorms can cover a large area and are typically made up of several individual storm cells. Individual cell motions move with the average 700mb wind direction

which tilts the storm as well as increases instability and the strength of the thunderstorms upward vertical motion we call the updraft. Potential severe conditions include possible flash floods, large hail, and short-lived tornadoes. Both large hail and tornadoes associated with multi-cell storms occur most often in the early afternoon with the first thunderstorms due to the aid of daytime heating.

Supercell

Supercells can be frontal or orographic. Orographic storms have an average duration of 1 to 3 hours and are associated with forced convection on the upstream side of mountain ranges. Frontal storms have an average duration of 4 to 8 hours and are commonly found on the warm side of the cold front. Like Multicell thunderstorms, a Supercell is made up of a very large storm cell that has been able to develop and increase intensity due to an abundance of each ingredient discussed earlier. Supercell thunderstorms are the most likely to produce the most severe weather conditions such as large hail, high damaging winds, and tornadoes.

Thunderstorm Intensities

There are three intensities or types of thunderstorms: standard, moderate, and severe.

Standard: Standard thunderstorms have present lightning within 5 miles of a location, with or without rain, surface winds less than 35KT, and no hail.

Moderate: Moderate thunderstorms contain the same thresholds as standard thunderstorms but have surface winds of 35-49KT, and/or hail less than 3/4 inch.

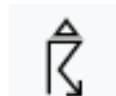
Severe: If surface winds reach 50KT or higher, or hail is present with a diameter of 3/4 inch or greater, the thunderstorm's intensity is severe. Tornadoic thunderstorms most often occur with moderate to severe thunderstorms.

Station Plot Symbols

Thunderstorm



Thunderstorms with Hail



Exercise / Practice: Thunderstorm Type and Intensity

Identifying Mesoscale Weather Features

Hail

Hail forms in the updraft (area of upward vertical motion enhancing lift) of a thunderstorm near the freezing level (the height of 0°C above the surface). Hail size will vary based upon the strength of the updraft. A strong updraft produces large hail while a weak updraft produces small

hail. Thunderstorm updrafts lift raindrops to freezing temperatures where internal thunderstorm winds push the raindrops between warmer and colder temperatures. The more these raindrops travel between above and below freezing altitudes, layers form on the hailstone causing them to increase in size. When the hailstones become too heavy for the updraft to keep aloft, they fall to the surface. Although possible with any type of thunderstorm, hail is most common with supercell thunderstorms and is usually associated with other severe weather features.

Squall Line

A squall line is a line of thunderstorms downstream of an inactive cold front. Their formation begins when 850-mb winds are faster than the surface winds behind the cold front. The upper winds push cold air downstream of the surface front creating a convergent zone on the surface with the southerly winds southeast of a low-pressure center creating a line of thunderstorms 50-150 miles ahead of the inactive cold front.

Tornadoes

Tornadoes are violently rotating columns of air that extend from a thunderstorm to the surface. Tornadoes can be extremely destructive and appear most commonly with severe thunderstorms. Ideal conditions involve mixing warm, moist air, cool, dry air, and enough wind shear (changing wind speed / direction) with height to create rotation within a thunderstorm's updraft. This rotation is then tilted vertically within the storm and eventually reaching the surface.

Tornado Types

- Rope - These are very thin and have a rope-like appearance.
- Cone - Shape is narrower on ground and wider at the cloud base. It is generally more dangerous than a rope tornado.
- Stovepipe - If the base and top have the same width it is a stovepipe tornado
- Wedge - The largest and most destructive tornado type. They are wider than they are tall with a width of 1/2 mile or greater.

Stages of Development:

- Initial - Majority of the tornado is invisible due to a lack of debris.
- Intensification - Debris extends from the surface to the cloud base. A condensation funnel cloud may be present.
- Mature - Tornado attains maximum width and it is very intense. Surface damage results from directional and speed shear.
- Dissipating - Tornado decreases diameter, tilts away from vertical, and its damage path decreases.
- Decaying - Tornado resembles a rope-like tube as the vortex lifts from the surface.

Intensities. We measure tornado intensity using the Enhanced Fujita Scale. It estimates tornado speeds based on a variety of damage indicators and ranging from visible damage to complete destruction. The Fujita scale has six levels ranging from EF0-EF5.

Thunderstorm Downburst

Thunderstorm downbursts typically occur in areas with higher levels of instability fueled by high surface temperatures, as well as the presence of an extremely dry layer of air in the mid-

levels around 700mb. This dry air increases cooling of falling precipitation, increasing its density, resulting in air rushing to the ground faster than the falling precipitation. Depending on the levels of instability and the intensity of the dry air intrusion, wind speeds from a downburst may be anywhere from 35KT to exceeding 100KT.

Exercise / Practice: Hail / Tornadoes / Downbursts

Heavy Precipitation

Heavy precipitation is classified as two or more inches of precipitation occurring within 12 hours and can be associated with or without thunderstorms.

Convective (associated with thunderstorms): Convective heavy precipitation occurs with slow-moving or stationary thunderstorms, severe thunderstorms, surface fronts, and embedded thunderstorms poleward of a warm front.

Non-convective (not associated with thunderstorms): Non-convective heavy precipitation most often occurs poleward of surface warm fronts, downstream of slow moving upper-level low-height centers, and with upslope winds on the windward side of mountains until winds decrease speed or change direction.

Station Plot Symbols

Rain

Light Rain



Moderate Rain



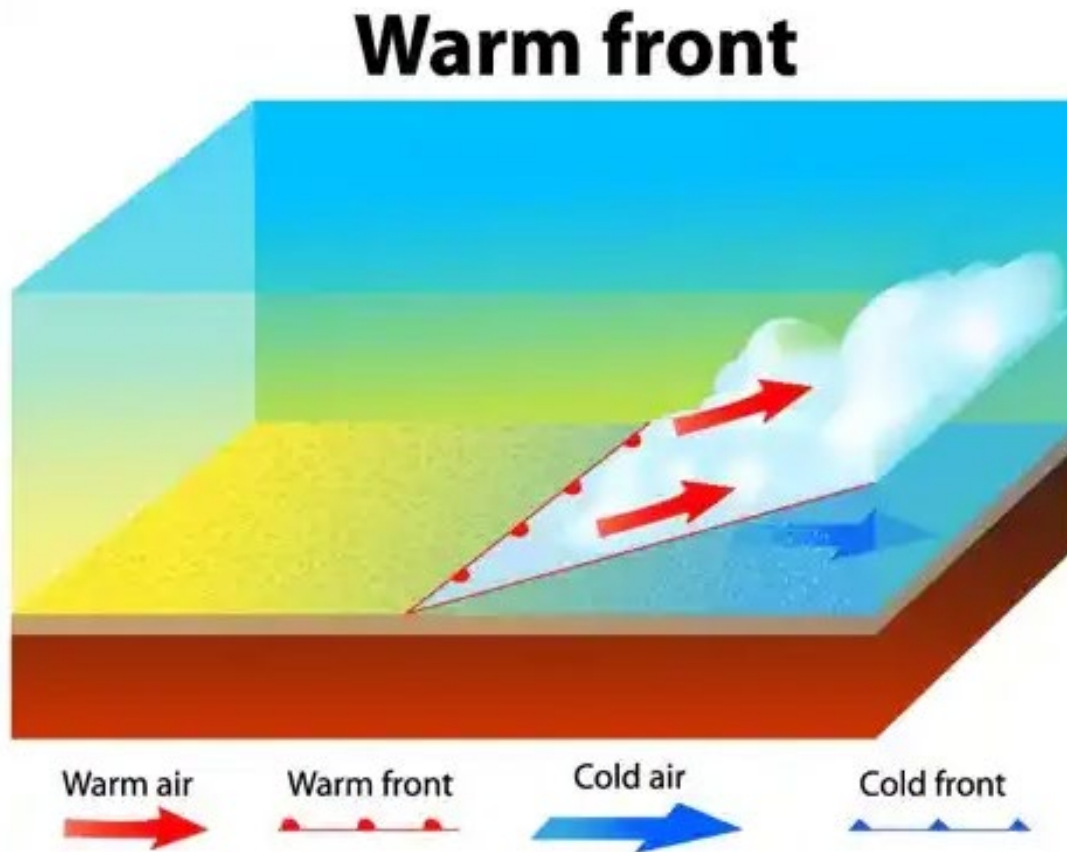
Heavy Rain



Freezing Precipitation

Freezing precipitation forms in areas where the upper levels are below freezing and falling precipitation passes through a warm layer (temperatures above freezing), then re-enter a cold layer (temperatures below freezing) just before hitting a surface that is also at or below freezing. Freezing precipitation occurs when the layer of freezing air close to the surface is shallow enough that the precipitation does not have enough time to completely freeze before reaching the surface. Instead, the water freezes on contact with the surface, creating a coating of ice on the surface.

Freezing precipitation is one of the most hazardous weather features and occurs most often poleward of surface warm fronts. The surface ahead of (poleward) of a warm front could be below freezing, while the rising warm air in a slope ahead of the front creates the warm layer aloft needed for freezing precipitation to occur.



Station Plot Symbol

Freezing Rain



Exercise / Practice: Heavy / Freezing Precipitation

Upslope and Downslope Winds

Upslope winds form when the surface synoptic flow is perpendicular to a mountain range. Surface winds move up the mountain increasing orographic lift causing clouds and potential precipitation or thunderstorms to develop on the windward (upstream) side of the mountain. Downslope winds require stable air near a mountaintop with the upper-level synoptic flow perpendicular to the mountain range. Upper-level winds push cold air down the lee side

(downstream) of the mountain where the warm air toward the base of the mountain is forced upward increasing upward vertical motion.

Sea Breeze and Land Breeze

Sea Breeze: A sea breeze forms along continental coastlines when surface winds blow from water to land. It is a daytime feature that reaches its peak in the afternoon. Land heats quicker than water during the day resulting in warmer air over the land and cooler air over the water. The warm air rises and cooler air from coastal waters replaces the warm air. Moist, onshore flow converges over land producing potential air mass thunderstorms in a line parallel to the coast 5-10 miles inland.

Land Breeze: Land breeze occurs when surface winds blow from the land to water. It is a nighttime feature reaching its peak in the early morning before sunrise. During the night, water retains heat longer than land leaving warmer air over the water. The warm air rises and cooler air over the land replaces the warm air. Offshore flow converges over water producing clouds and potential air mass thunderstorms over the water near the shoreline.

Exercise / Practice: Upslope/Downslope Winds / Land/Sea Breeze

Mist and Fog

Mist and fog form when air near the surface becomes saturated, reducing visibility due to suspended water droplets. They are typically early morning features, often developing overnight and dissipating after sunrise. Fog typically occurs when air parcels become fully saturated due to recent precipitation, moist air advection, or upslope winds. Mist forms under similar conditions but with less restriction to visibility. Mist and fog are common in areas under high atmospheric stability, with winds less than 10 knots, and clear skies.



Station Plot Symbols

Mist



Fog



Exercise / Practice: Mist and Fog

Exercise / Practice: Mesoscale Analysis

PC: Mesoscale Analysis